Smaller Diophantine forms of ZFC consistency

Combining the blank-input compiler, ordinary Pell ranks,
and signed auxiliary coordinates

Research contribution prepared for the ProveIt repository

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Abstract

We combine three existing construction techniques in ProveIt and prove additional domain reductions. With the fixed coefficients of package `06-blank67`, an ordinary strong Pell equation gives a system of six equations in fifteen positive unknowns, of maximum degree 28, evaluated in 61 binary arithmetic operations. This improves the previous 16-unknown, seven-equation, degree-28, 63-operation system, and lowers the degree of the previous 15-unknown, six-equation, 61-operation system. The corresponding product polynomial has exact degree 99 and 69 operations; its sum-of-squares alternative has exact degree 56 and 78 operations. A quadratic lift uses 33 positive unknowns and 24 equations, giving a single quartic in 33 positive unknowns.

A separate argument removes positivity from six auxiliary coordinates, permits an odd strong multiplier, and makes the power scale a square. Only seven coordinates still need a positive parametrization. This gives explicit equations in 29 unrestricted integer unknowns, including a 121-operation polynomial of exact degree 188. Two alternative integer polynomials use 119 and 120 operations at degrees 300 and 236.

For every displayed formulation, ZFC proves that absence of solutions in its stated domain is equivalent to Con(ZFC). The proof uses the repository's ordinary K215 calibration and native-window/Pell compiler proofs; these dependencies are identified explicitly. The new bridge is not a Lean or Rocq formalization. The four coefficients are closed finite integer recipes, with $B = 2^{5^{67}}$; their binary expansions are not materialized. No global minimum, consistency proof, or improvement to a varying-input universal operation bound is claimed.

Contents

1 Results and the meaning of simplicity **3**

2 What the eight earlier packages contribute **3**

3 The six equations and a single polynomial **4**

4 Pell preliminaries and the ordinary rank lemma **5**

5 Soundness, with the proof order made explicit **6**

5.1 Unit factors and the zero-scale boundary 6

5.2 Bounds before decoding any mask 7

5.3 First and main Pell indices 7

5.4 Signed auxiliary congruences recover the exact index 7

5.5 No wrap, and the two remaining signs 8

5.6 The sign of the main root and power recovery 8

5.7 The retained computation compiler 9

6 Completeness and the ZFC equivalence **9**

7 Twenty-nine unrestricted integer unknowns **10**

7.1 An odd strong multiplier is sufficient 10

7.2 The power scale can be a square 10

7.3 The last seven positive coordinates 10

8 A quartic in thirty-three positive unknowns **11**

9 Exact degrees and reproducible operation counts **12**

10 Verification, provenance, and limits of the claim **13**

11 Further research questions **13**

A A closed specification of the four coefficients **14**

A.1 The local alphabet and the two distinguished windows 14

A.2 Numerical layout 15

A.3 Why this recipe encodes a blank halting computation 16

B Reproduction and artifact map **16**

1 Results and the meaning of simplicity

The user-specified objective has several different size measures: number of unknowns, number of equations, degree, arithmetic circuit length, and coefficient height. There is no agreed total ordering of these measures. We give explicit improvements within the supplied compiler family and preserve the alternatives needed to compare them fairly. Our recommended certificate is the six-equation system below. For a single equation in positive integers, the new degree-99 product is a useful degree/operation compromise. For unrestricted integers, the 29-variable degree-188 product is the main new formulation.

All operation counts charge each binary $+$, $-$, and multiplication, including multiplication by a fixed numeral. Fixed coefficients and reuse of a computed register are free. A system's equality comparisons are not arithmetic operations. Conversion to a single sum of squares charges its subtractions, squares, and additions in full. These are upper bounds for specified circuits, not lower bounds for all circuits.

New presentation	Unknowns	Equations	Degree	Operations
Ordinary Pell system	15 positive	6	28 max.	61
Product P_{69}	15 positive	1	99	69
Sum of squares S_{78}	15 positive	1	56	78
Quadratic lift	33 positive	24	2 max.	—
Quartic Q_{33}	33 positive	1	4	—
Integer system	29 integers	6	50 max.	113
Integer product	29 integers	1	188	121
Integer sum of squares	29 integers	1	100	130
Integer normalized alternative	29 integers	1	300	119
Integer reduced alternative	29 integers	1	236	120

All degrees in the table are exact, including the maximum residual degrees. The artifacts give every arithmetic gate and every input port.

Theorem 1.1 (Main consistency formulations)**.** *Use the four fixed coefficients in Appendix A, bound to the exact K215 table in the archive. Let $\mathcal{S}$ be the system in Section 3, let P_{69} and S_{78} be its displayed single-polynomial forms, and let Q_{33} and $\widehat{P}_{121}$ be the quartic and unrestricted-integer formulations below. Then ZFC proves each of the equivalences*

$$\begin{aligned}
\exists \mathbf{x} \in \mathbb{Z}_{>0}^{15}\ \mathcal{S}(\mathbf{x}) &\longleftrightarrow \exists \mathbf{x} \in \mathbb{Z}_{>0}^{15}\ P_{69}(\mathbf{x}) = 0 \\
&\longleftrightarrow \exists \mathbf{x} \in \mathbb{Z}_{>0}^{15}\ S_{78}(\mathbf{x}) = 0 \\
&\longleftrightarrow \exists \mathbf{v} \in \mathbb{Z}_{>0}^{33}\ Q_{33}(\mathbf{v}) = 0 \\
&\longleftrightarrow \exists \mathbf{t} \in \mathbb{Z}^{29}\ \widehat{P}_{121}(\mathbf{t}) = 0 \\
&\longleftrightarrow \exists p\ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot).
\end{aligned}$$

Consequently, the negation of each solution predicate is provably equivalent in ZFC to Con(ZFC).

Here $\mathrm{Proof}_{\mathrm{ZFC}}$ is a standard effective finite proof predicate, or the equivalent explicit axiom presentation used by the repository's calibration. Standard finite proof translations between presentations are formalizable in ZFC.

2 What the eight earlier packages contribute

The comparison is pinned to `3b201cdea32d2c816f5f3d20734051f25fd3ef14`. The eight packages themselves share the earlier snapshot `cdd582bec13f908df3711fa9e05d1ddac368a5db`. The

repository reconciliation records their differing coefficients, proof status, and metrics [1]. The following table selects the constructions relevant to the present improvement; it is not a claim to reproduce every prior artifact.

Package	Relevant contribution	Selected prior size
01-explicit82	Direct fixed-input specialization with retained witnesses	18 vars., 82 ops.
02-system63	Ordinary strong-root equation and reduced auxiliary coefficient	16/7/28/63
03-system64	Equivalent fixed-marker system; strong-root elimination	16/7/28/64
04-poly73	Affine marker guard and two-sided normalization	16 vars., 73 ops.
05-fixed15-16	Witness restoration and strong-root elimination alternatives	15 vars., 80 or 82 ops.
06-blank67	Raw K215, no input interface, numeric coefficient recipe	15/6/60/61; P_{67}
07-fixed70	Fixed input with altered Pell scale	17/8/28/61
08-start-only67	Independent input-free arithmetic; quadratic register elimination	34 vars., 25 quadratics

In four-number entries the order is unknowns/equations/degree/operations. Packages 06 and 08 independently have the same 15-variable, 67-operation, degree-155 polynomial skeleton, but different coefficient integers. We retain *only package 06's coefficient recipe*; borrowing the ordinary-root algebra of 02 or the register eliminations of 08 does not authorize substituting their different coefficients.

The new system improves the 16/7/28/63 system on unknowns, equations, and operations, while retaining degree 28. It improves the 15/6/60/61 system on degree without increasing any other listed size. The new quartic reduces the earlier 34 positive variables and 25 quadratic relations to 33 and 24. The degree-99 product costs two operations more than P_{67}; we do not describe it as beating the operation record. The earlier degree-123 product costs 68 operations. The new degree-56 sum of squares costs 78 operations, versus 83 for package 02's 16-variable degree-56 form and 79 for package 06's degree-92 form.

Fifteen is not the least known possible number of positive/natural unknowns in general Diophantine representation. Jones's classical nine-unknown construction has a very different degree and description cost [9]. Similarly, 29 is a size bound for these fully specified integer equations, not a universal minimum over all integer representations; see also [10]. The present gain is a concrete combination and proof simplification relative to the supplied artifacts.

3 The six equations and a single polynomial

Let β, K, M_C, M_F^* be the fixed coefficients of Appendix A. In particular, $\beta = B - 1$, and M_F^* is the *shifted* source coefficient $M_F + B - 1$. Supply exactly the fifteen positive integers

$$(J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma).$$

Define polynomial abbreviations

$$\begin{aligned} &q = \beta J + 1, && C = q - F - \alpha, && X = wq, && Y = sq^3, \\ &E = XY, && k = \eta + \zeta, && c = kY + \eta, && a = E + Y, \\ &H = 4a + 3, && \Delta = a^2 + H = (a+2)^2 - 1, && D = X + ac + \gamma H. \end{aligned} \tag{1}$$

Further put

$$\begin{aligned} G &= (q-1)(q-F)+\alpha,\\ R &= G(q^2-1)+(M_C+qM_F^*)J,\\ L &= EkY, \qquad\qquad V = c(Tf-1)-Rf^2, \quad I = ic^2, \quad U = \Delta(f^2-1). \end{aligned} \tag{2}$$

These are substitutions, not additional unknowns or constraints. All exponents of unknowns are fixed nonnegative integers. Set

$$\begin{aligned} N_0 &= \tau^2 - L(L+k), & N_1 &= D^2-\Delta c^2,\\ N_a &= U(V^2-y^2)+y^2, & N_k &= k-hE-R,\\ N_t &= (K+w)C+q-F-z(q-1), & N_s &= 1+I^2-U. \end{aligned} \tag{3}$$

Our recommended system is

$$\boxed{N_0 = N_1 = N_a = N_k = N_t = 1, \qquad I^2 = U.} \tag{S}$$

The degree-99 single equation is

$$\boxed{P_{69} = N_0N_1N_aN_kN_tN_s - 1 = 0.} \tag{P}$$

The degree-56 alternative is

$$\boxed{S_{78} = (N_0-1)^2+(N_1-1)^2+(N_a-1)^2+(N_k-1)^2+(N_t-1)^2+(I^2-U)^2 = 0.} \tag{SOS}$$

This is an ordinary polynomial problem: the coefficient recipe has no variable program parameter, hidden exponential relation, extraction function, or variable-length equation list.

4 Pell preliminaries and the ordinary rank lemma

For an integer $A \geq 2$, write

$$\chi_A(n) + \psi_A(n)\sqrt{A^2-1} = (A+\sqrt{A^2-1})^n \qquad (n \in \mathbb{N}).$$

Both sequences are integer recurrences with characteristic equation $t^2 - 2At + 1 = 0$. Positive solutions of $d^2-(A^2-1)c^2 = 1$ have $(d,c) = (\chi_A(p), \psi_A(p))$. We use the addition and composition identities, the growth bounds

$$(2A-1)^{n-1} \leq \psi_A(n) \leq (2A)^{n-1},$$

for $n \geq 1$, and strong divisibility $\gcd(\psi_A(u), \psi_A(v)) = \psi_A(\gcd(u,v))$. Strong divisibility follows from the addition formula, $\gcd(\chi_A(n), \psi_A(n)) = 1$, and the Euclidean algorithm on indices. These assertions concern integer-coefficient units; no assertion about the full ring of algebraic integers is needed.

Lemma 4.1 (Ordinary strong rank)**.** *Let $\Delta = A^2-1$, $c = \psi_A(p)$, and $c > A\Delta^2$. If nonzero integers i, f satisfy*

$$(ic^2)^2 = \Delta(f^2-1),$$

then there is $m > 0$ with

$$|f| = \chi_A(m), \qquad p \mid m, \qquad c \mid m, \qquad |i|c^2 = \Delta\psi_A(m).$$

In particular $c^2 \mid f^2-1$.

Proof. We include the inherited rank argument [5] because it is the essential justification for the change of strong equation. Write $\Delta = d_0 s_0^2$, with $d_0 > 1$ squarefree, and let $F_0 + y_0\sqrt{d_0}$ be the least integer-coefficient norm-one unit greater than 1. Division by successive powers of this least unit proves that all positive integer solutions in this field are its powers. Thus for some $e \geq 1$, with $S = \psi_{F_0}(e)$,

$$A = \chi_{F_0}(e), \quad s_0 = y_0 S, \quad \Delta = (F_0^2 - 1)S^2, \quad \psi_{F_0}(ep) = Sc.$$

Set $\rho = |i|c^2$. The equality $\rho^2 = \Delta(f^2 - 1)$ first gives $s_0 \mid \rho$, then $d_0 \mid \rho/s_0$ by squarefreeness. Hence $|f|^2 - d_0(\rho/(d_0 s_0))^2 = 1$, and for some $n > 0$

$$|f| = \chi_{F_0}(n), \qquad \rho = (F_0^2 - 1)S\psi_{F_0}(n).$$

Let $T_0 = (F_0^2 - 1)S \leq \Delta$ and $g = \gcd(n, ep)$. Strong divisibility, together with $c^2 \mid T_0\psi_{F_0}(n)$, gives

$$c \leq \gcd(c^2, \Delta c) \leq T_0\psi_{F_0}(g), \qquad \psi_{F_0}(g) \geq c/\Delta.$$

If $g < ep$, then $2g \leq ep$. Since $\psi_{F_0}(2g) > 2\psi_{F_0}(g)^2$ and $S < A$, we obtain $\psi_{F_0}(g)^2 < Ac/2$. This contradicts $c^2/\Delta^2 > Ac$. Therefore $ep \mid n$; writing $n = em$ gives $p \mid m$, $|f| = \chi_A(m)$, and $\rho = \Delta\psi_A(m)$.

Write $m = pb$ and $d = \chi_A(p)$. Expanding $(d + c\sqrt{\Delta})^b$ modulo c^2 gives $\psi_A(pb)/c \equiv bd^{b-1} \pmod{c}$. Thus $c \mid \Delta b$. Also the binomial expansion modulo Δ gives $c \equiv pA^{p-1} \pmod{\Delta}$, so $\gcd(c, \Delta) = \gcd(p, \Delta)$ divides p. Consequently $c \mid pb = m$. Finally $f^2 - 1 = \Delta\psi_A(m)^2 = |i|c^2\psi_A(m)$. □

Lemma 4.2 (Two unit exclusions)**.** *If $v > 1$ is an integer, $x^2 - v(v+1)t^2 = -1$ has no integer solution. If $\Delta = A^2 - 1$, neither $D^2 - \Delta c^2$ nor $1 + (ic^2)^2 - \Delta(f^2 - 1)$ can equal -1.*

Proof. For the first assertion take a solution with $t > 0$ minimal and $x > 0$. Its equation implies $vt < x < (v + \frac{1}{2})t$. Multiplication by $2v + 1 - 2\sqrt{v(v+1)}$ gives another integer solution whose coefficient $t' = (2v+1)t - 2x$ satisfies $0 < t' < t$, a contradiction. For the others, Δ is 0 or 3 modulo 4. The first expression cannot be 3 modulo 4. For the second, note that $\Delta(f^2 - 1)$ is 0 or 1 modulo 4; subtracting it from a square cannot give -2 modulo 4. □

5 Soundness, with the proof order made explicit

We give a proof that also covers the signed-domain extension. Initially require $J, F, \alpha, z, s, \eta, \zeta > 0$, $w \geq 0$, and $i \neq 0$. Allow f, h, T, τ, y, γ to be arbitrary integers. For now assume that either (S) or (P) holds.

5.1 Unit factors and the zero-scale boundary

At a product zero every factor is an integer unit. The mod-4 arguments give $N_1 = N_s = 1$. Once $N_s = 1$, $U = I^2$ is a square and N_a is congruent either to y^2 or to V^2 modulo 4; therefore $N_a = 1$. For $X > 0$, the first norm is

$$N_0 = \tau^2 - v(v+1)k^2, \qquad v = XY^2 > 1,$$

so $N_0 = 1$ by Lemma 4.2. If $X = 0$, it is simply τ^2 and again cannot be -1. Thus $N_k = \varepsilon$, $N_t = \nu$, where $\varepsilon, \nu \in \{1, -1\}$ and $\varepsilon\nu = 1$.

If $w = 0$, then $X = 0$, $a = Y > 0$, $D = ac + \gamma H$, and $\Delta = a^2 + H$. The main norm reduces modulo H to

$$D^2 - \Delta c^2 \equiv 0 \pmod{H},$$

contradicting its value 1 because $H = 4Y + 3 > 1$. Hence $w > 0$. This short modular argument is the reason a square parametrization of w does not require a separate nonzero constraint.

5.2 Bounds before decoding any mask

Since $z, F \geq 1$,

$$(q-F) - z(q-1) = 1 - F - (z-1)(q-1) \leq 0.$$

With $K + w \geq 2$, $N_t \in \{1, -1\}$ therefore forces $C \geq 0$. It does *not* yet force $C > 0$. It follows that $q - F = C + \alpha > 0$ and $1 \leq F \leq q-1$. Using the authentic masks $0 < M_C, M_F < B - 1$ and $M_F^* = M_F + B - 1$, one gets

$$\begin{aligned} q \leq G = q\alpha + (q-1)C \leq q^2 - q, \\ 0 < (M_C + qM_F^*)J < 2q^2 - q - 1, \\ q(q^2-1) < R < q^4 - q^3 + q^2 - 1 < q^4 - 2, \\ q \geq B \geq 16, \quad X \geq q, \quad Y \geq q^3, \quad E \geq q^4 > R + 2. \end{aligned} \tag{4}$$

In particular $R > 3q + 1$ and $a > R + 2$. None of these conclusions uses a dyadic value of q, an input equation, or $C > 0$.

5.3 First and main Pell indices

Put $A = a + 2$ and $P_0 = 2XY^2 + 1$. The least integer-coefficient norm-one unit greater than 1 for $v(v+1)$ is $2v + 1 + 2\sqrt{v(v+1)}$: coefficient 1 cannot solve the norm, and coefficient 2 does. Consequently

$$k = 2\psi_{P_0}(n), \quad |\tau| = \chi_{P_0}(n), \qquad c = \psi_A(p), \quad |D| = \chi_A(p)$$

for $n, p \geq 1$. Since $P_0 \equiv 1 \pmod{E}$,

$$2n = R + \varepsilon + v_1 E, \qquad v_1 \geq 0, \qquad n \geq (R-1)/2 \geq 24. \tag{5}$$

The nonnegativity of v_1 uses $0 < R + \varepsilon < E$. Since $P_0 > A$ and $c > k$, monotonicity of the Pell sequences gives $p > n$. Their growth then gives

$$c > A\Delta^2, \qquad c > 2p.$$

Even when h was allowed to have either sign, $k = R + \varepsilon + hE > 0$ and (4) imply $h \geq 0$. Thus $c > kY \geq (R-1)Y > 2R$. In fact $h = 0$ is impossible: it would give $k = R \pm 1$; $n = 1$ gives $k = 2$, whereas $n \geq 2$ gives $k \geq 4P_0 > R + 1$. We may therefore recover $h > 0$, although the weaker bound already suffices.

5.4 Signed auxiliary congruences recover the exact index

Lemma 4.1 now applies, before any sign restriction on V or T is used. Write $f_0 = |f|$ and $\rho = |i|c^2 > 1$. We obtain

$$f_0 = \chi_A(m), \quad c \mid m, \quad m \geq c > 2p, \quad \rho = \Delta\psi_A(m), \quad c^2 \mid f^2 - 1.$$

The auxiliary norm is

$$\rho^2 V^2 - (\rho^2 - 1)y^2 = 1.$$

It forces $V \neq 0$ and $y \neq 0$. Pell classification gives $\rho|V| = \chi_\rho(\ell)$ and $|y| = \psi_\rho(\ell)$. The index ℓ is odd, since an even-index χ_ρ is ± 1 modulo ρ. Write $\ell = 2u + 1$ and $\sigma = \operatorname{sgn}(V)$. The integer polynomials Q_u determined by

$$\chi_Z(2u+1) = ZQ_u(Z^2)$$

satisfy

$$Q_u(0) = (-1)^u(2u+1), \qquad Q_u(1 - A^2) = (-1)^u\psi_A(2u+1).$$

For example $Q_0 = 1$, $Q_1 = 4t - 3$, and $Q_{u+2} = (4t-2)Q_{u+1} - Q_u$; the identities follow by recurrence.

The definition of V supplies, without any auxiliary positivity proof,

$$V \equiv -c \pmod{f_0}, \qquad V \equiv -R \pmod{c}.$$

The second congruence uses the rank conclusion $c^2 \mid f^2 - 1$; it must not be obtained by illicit cancellation of Δ in the ordinary strong equation. Since $\rho^2 \equiv 1 - A^2 \pmod{f_0}$ and $c \mid \rho$, we have

$$\sigma(-1)^u \psi_A(\ell) \equiv -c \pmod{f_0}, \qquad \sigma(-1)^u \ell \equiv -R \pmod{c}.$$

Squaring the first congruence and using $\chi_A(2j) = 1 + 2\Delta\psi_A(j)^2$ yields

$$\chi_A(2\ell) \equiv \chi_A(2p) \pmod{\chi_A(m)}.$$

The Pell step-down lemma [6], applicable to $0 < 2p < m$, gives $\ell \equiv \pm p \pmod{m}$. A short proof reduces indices using $\chi_A(j+2m) \equiv -\chi_A(j) \pmod{\chi_A(m)}$ and evenness, and distinguishes representatives by strict monotonicity and $2\chi_A(m-1) < \chi_A(m)$. As $c \mid m$, the other congruence implies $R \equiv \pm p \pmod{c}$. The bounds $0 < R, p < c/2$ force

$$\boxed{p = R.}$$

This is the key proof simplification: it eliminates the need to restore positive quotient coordinates before identifying the index.

5.5 No wrap, and the two remaining signs

If $v_1 \geq 1$ in (5), then $2n \geq R - 1 + E > 2R$, contradicting $n < p = R$. Hence $2n = R + \varepsilon$. If $\varepsilon = -1$, then $p = 2n + 1$. With $Q = \chi_A(2) > P_0$, duplication and monotonicity imply

$$\psi_A(2n) = 2A\psi_Q(n) \geq 2A\psi_{P_0}(n) = Ak.$$

This contradicts $c = \psi_A(2n+1) < k(Y+1)$, since $A > Y + 1$. Therefore $\varepsilon = \nu = 1$, $R = 2n - 1$ is odd, and $N_t = 1$ now gives $C > 0$. Every product zero satisfies the six equations individually. The converse is immediate, proving their equality of zero sets on the positive domain.

5.6 The sign of the main root and power recovery

Put $r_0 = (R-1)/2 \geq 2$. The inherited lower Pell-ratio estimate uses only $c = \psi_A(R)$ and $k = 2\psi_{P_0}(r_0 + 1)$:

$$\frac{c}{k/2} > \frac{(X+1)^{2r_0}}{X^{r_0}} > X^{r_0}.$$

For completeness, the stronger first comparison follows by bounding the numerator below by $(2A-1)^{2r_0}$ and the denominator above by $(2P_0)^{r_0}$; the extra factor is greater than one because $6XY^2 > a$. Together with $c/k < Y + 1$, it gives

$$Y > X^{r_0}/3, \qquad a > X^{r_0+1}/3.$$

As $X \geq 16$ and $r_0 \geq 2$, $(X/4)^{r_0} > 6$, and hence

$$0 < X2^R = 2X4^{r_0} < a < H - 1. \tag{6}$$

The recurrence gives $\chi_A(R) - a\psi_A(R) \equiv 2^R \pmod{H}$. Also $(\chi_A(R) - ac)(\chi_A(R) + ac) = 1 + Hc^2$. If $D = -\chi_A(R)$, the defining relation $D - ac = X + \gamma H$ would imply $X2^R \equiv -1 \pmod{H}$,

contradicting (6). Thus $D > 0$ even for signed γ. We conclude $X \equiv 2^R \pmod{H}$; both numbers lie strictly between zero and a, so

$$\boxed{X = 2^R.}$$

Since $q \mid X$, $q = 2^t$. The relation $q \equiv 1 \pmod{2^d - 1}$ gives $d \mid t$ by the order-of-two argument; write $q = B^N$. The bound $R > 3q$ implies $q^3 \mid X$. Thus the post-index ratio and binomial projection calculation in Sections 5–6 of `pell_kernel_half_binomial42.md` applies to the recovered positive X, Y, c, k, with $p = R$, $2n = R + 1$, and the strict ratio $kY < c < k(Y + 1)$. This calculation does not use the signs of the remaining auxiliary coordinates. We reuse that part of the proof, not its full positive-witness theorem on a signed tuple.

5.7 The retained computation compiler

At this point the inherited binomial calculation gives $Y = \lfloor (X + 1)^{R-1} / X^{(R-1)/2} \rfloor / 2$ and $\nu_2(Y) = \mathrm{pc}(R) - 2$. The paid factor $q^3 \mid Y$ supplies the population threshold used by the inverse mask theorem. The resulting relations are

$$(C - 1) \,\&\, (M_C J + 1) = 0, \qquad F \,\&\, (M_F J - 1) = 0.$$

Here & is used only in this proof, not in the polynomial. The carry-free convolution and anchor argument of the retained compiler extract a cyclic word of local windows with Start at the origin and the distinguished impossible selector excluded. Start's top/center mismatch rules out zero temporal shift. The blank-window lemma extracts a finite halting computation of the literal K215 table. The exact compiler proof dependencies and the local language are included in the archive; Appendix A fixes all coefficients. This inherited decoding stage is unchanged by the ordinary-root or sign substitutions.

6 Completeness and the ZFC equivalence

Suppose K215 has a finite blank halting run. The retained compiler pads it to a helical rectangle with spatial width and temporal height both large powers of five. It constructs the outer coordinates and the first and main Pell witnesses for the actual packed index R. The dummy-bit adjustment arranges the required temporal congruence at that same R; it is not an earlier provisional index. In particular

$$q = B^N, \quad N \text{ a power of five}, \quad X = 2^R, \quad R \equiv 3 \pmod{4},$$

and all originally positive coordinates can be chosen positive. The already-proved normalized construction has $f^2 - 1 = \Delta(i_{\mathrm{norm}} c^2)^2$. Set $i_{\mathrm{ord}} = \Delta i_{\mathrm{norm}}$. Then

$$(i_{\mathrm{ord}} c^2)^2 = \Delta(f^2 - 1),$$

and the auxiliary coefficient and all other equations are unchanged on the solution. This proves completeness of (S). It proves equivalence of *existence* predicates, rather than identity of tuples with the same strong multiplier.

The existing K215 calibration states

$$\mathrm{Halt}(\mathrm{K215}, \mathrm{blank}) \iff \exists p \; \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot).$$

Its ordinary proof, the finite syntax and computation encodings, the Pell and coefficient arguments, and the present transformations are formalizable in ZFC. Composing these arguments proves Theorem 1.1. This reasoning does not infer ZFC-provability from the metatheoretic soundness of Lean or Rocq.

7 Twenty-nine unrestricted integer unknowns

The signed soundness proof above already established more than positive-domain equivalence. It permits arbitrary integer values for

$$f, h, T, \tau, y, \gamma,$$

provided $i \neq 0$, $w \geq 0$, and $J, F, \alpha, z, s, \eta, \zeta > 0$. We next obtain convenient polynomial parametrizations of i and w without losing completeness.

7.1 An odd strong multiplier is sufficient

In a canonical halting witness, $A = a + 2$ is even and R is odd; therefore $c = \psi_A(R)$ is odd. Choose the auxiliary index

$$m = cR$$

instead of the older $2cR$. Expanding $(\chi_A(R) + c\sqrt{\Delta})^c$ shows $c^2 \mid \psi_A(cR)$. Since m is odd and $\psi_A(m) \equiv m \pmod 2$, the integer

$$i_{\text{norm}} = \psi_A(cR)/c^2$$

is positive and odd. Also Δ is odd, so $i_{\text{ord}} = \Delta i_{\text{norm}}$ is positive and odd. Set $f = \chi_A(m)$ and $\rho = \Delta\psi_A(m)$, and take

$$V = \chi_\rho(R)/\rho, \qquad y = \psi_\rho(R).$$

The quotient is an integer because R is odd. As $R \equiv 3 \pmod 4$, the identities for Q_u give $V \equiv -c \pmod f$ and $V \equiv -R \pmod c$. The coprimality $\gcd(c, f) = 1$ and $f^2 \equiv 1 \pmod c$ make

$$T = \frac{V + c + Rf^2}{cf}$$

a positive integer. Thus the whole auxiliary block, not merely its strong equation, has a witness with odd i. We may replace i everywhere by $2r + 1$, with $r \in \mathbb{Z}$. It is never zero, and covers the canonical positive odd choices.

7.2 The power scale can be a square

The fixed $d = 5^{67}$ and canonical N are odd, and R is odd. Moreover $R > dN$. Therefore

$$w = X/q = 2^{R-dN} = \left(2^{(R-dN)/2}\right)^2.$$

We may replace w by u^2, $u \in \mathbb{Z}$. Soundness at $u = 0$ was already excluded by the main norm modulo H; no extra equation or inequality is needed.

7.3 The last seven positive coordinates

Define

$$\text{Pos}(a, b, c) = 1 + a^2 + b^2 + c(c + 1) \qquad (a, b, c \in \mathbb{Z}).$$

It is always positive. Conversely every $n > 0$ is represented: $4(n - 1) + 1$ is not of the excluded three-square form $4^e(8t + 7)$, and is 1 modulo 4. Legendre's three-square theorem therefore gives exactly one odd square and two even squares. Writing them as $(2c + 1)^2, (2a)^2, (2b)^2$ yields the formula.

For each $v \in \{J, F, \alpha, z, s, \eta, \zeta\}$ use a separate triple (v_1, v_2, v_3) and substitute

$$v = \text{Pos}(v_1, v_2, v_3), \qquad i = 2r + 1, \qquad w = u^2. \tag{7}$$

Keep f, h, T, τ, y, γ as unrestricted integers. There are exactly $7 \cdot 3 + 2 + 6 = 29$ unknowns. Applying these substitutions to P_{69} defines the explicit polynomial $\widehat{P}_{121}$. Its lack of an integer zero is provably equivalent in ZFC to Con(ZFC).

The substitutions cost $7 \cdot 7 + 2 + 1 = 52$ operations: each positive parametrization costs three multiplications and four additions, the odd affine form costs two, and the square costs one. Thus P_{69} becomes a 121-operation polynomial; the system becomes 113 operations; the sum of squares becomes 130 operations. The same signed proof applies to the normalized P_{67} and reduced P_{68} of package 06, giving 119- and 120-operation 29-integer alternatives. For normalized rank one may use the shorter direct argument $\psi_A(m) = |i|c^2 \Rightarrow pc \mid m$; the ordinary relaxed-rank lemma is then unnecessary.

8 A quartic in thirty-three positive unknowns

For a low-degree presentation, retain the fifteen original positive unknowns and introduce the following eighteen positive registers:

$$(n_3, E, p_0, c, a, X, D, c_2, G, \ell_2, R, f_2, v_0, V, g, I, b_0, \Delta).$$

In the equations below $q = \beta J + 1$, $k = \eta + \zeta$, and $H = 4a + 3$ are linear abbreviations, not additional unknowns. Take the following 24 quadratic equations:

$$\begin{aligned}
\ell_2 &= q^2 - 1, & n_3 &= (\ell_2 + 1)q,\\
X &= wq, & a &= E + n_3 s,\\
E &= X(a - E), & c &= \eta + (a - E)k,\\
p_0 &= E(c - \eta) + k, & \tau^2 &= p_0(p_0 - k) + 1,\\
D &= X + ac + \gamma H, & D^2 &= \Delta c_2 + 1,\\
\Delta &= a^2 + 4a + 3, & c_2 &= c^2,\\
k &= Eh + R + 1, & G &= (q-1)(q-F) + \alpha,\\
& & R &= G\ell_2 + (M_C + qM_F^*)J,\\
(K + w)(q - F - \alpha) &+ q - F - z(q-1) = 1,\\
f_2 &= f^2, & v_0 &= Tf - 1,\\
V &= cv_0 - Rf_2, & g &= y^2 - V^2,\\
I &= ic_2, & b_0 &= I^2,\\
b_0 &= \Delta(f_2 - 1), & y^2 - b_0 g &= 1.
\end{aligned} \tag{8}$$

Every displayed polynomial has degree at most two after inlining the three linear abbreviations. Let $r_1, \ldots, r_{24}$ be the left sides minus the right sides. Then

$$\boxed{Q_{33} = \sum_{j=1}^{24} r_j^2 = 0}$$

has exact degree four; the coefficient of τ^4 is 1.

To verify equivalence, the first equations force $n_3 = q^3$, $a - E = sq^3 = Y$, $E = XY$, and $c = kY + \eta$. The register $p_0 = EkY + k = L + k$ gives the original first norm. All other substitutions recover (S) directly. For the converse, a positive system solution has positive values for all these registers. The only nontrivial ones are v_0, V, g. Lemma 4.1 and $c > A\Delta^2$ give $f > 2c$ and $f^2 > \Delta + c$. Since $R < a < \Delta$,

$$U - (Rf^2 + c) = (\Delta - R)f^2 - \Delta - c > 0.$$

If $|V| > 1$, the auxiliary norm implies $|V| \geq 2U - 1$; but $T > 0$ gives $V > -Rf^2 - c > -U$. The values $V = \pm 1$ are excluded by $V \equiv -c \pmod{f}$ and $2 < c < f/2$. Thus $V > 1$, $v_0 = Tf - 1 > 0$, and the auxiliary norm gives $g = y^2 - V^2 > 0$. This proves a positive inverse lift.

Relative to the 34-variable lift of package 08, the removed register was the scaled quantity Δc^2. Its two defining norm equations are replaced by $D^2 = \Delta c_2 + 1$, and the strong register becomes $I = ic_2$. These are algebraic changes; package 06's coefficients remain the ones used here. Substituting $v = n_v + 1$ for each of the 33 positive unknowns gives a degree-four equation over $\mathbb{N}^{33}$. It is not legitimate simply to read $Q_{33} = 0$ over unrestricted integers without imposing its domain.

9 Exact degrees and reproducible operation counts

The ordinary system replaces four gates of the normalized system by four gates of the same cost:

Old gates	New gates
$i(\Delta c^2)$	$f^2 - 1$
$(i\Delta c^2)^2$	$\Delta(f^2 - 1)$
Δf^2	ic^2
$\Delta f^2 - (i\Delta c^2)^2$	$(ic^2)^2$

Each column costs three multiplications and one subtraction; the new last two registers are compared as $I^2 = U$. The result remains $61 = 34M + 27A$. Forming N_s costs two operations, and the product and subtraction of 1 cost six, giving $69 = 39M + 30A$. Six residual subtractions, six squares, and five additions give $78 = 40M + 38A$ for the sum of squares.

For the degree calculation put $Q = \beta J$, $k_0 = \eta + \zeta$, and $C_1 = Q - F - \alpha$. The exact factor degrees of $(N_0, N_1, N_a, N_k, N_t, N_s)$ are

$$(22, 18, 28, 7, 2, 22).$$

The only substantial top-degree cancellation occurs in N_1:

$$N_1 = 2ac(X + \gamma H) + (X + \gamma H)^2 - Hc^2.$$

Its leading term is $8\gamma a_{\mathrm{top}}^2 c_{\mathrm{top}}$. The product's full leading homogeneous form is

$$8Q^{55}h\gamma i^2 k_0^9 w^7 s^{16} T^2 f^4 (wC_1 - zQ). \tag{9}$$

In particular $J^{56}h\gamma i^2\eta^9 w^7 s^{16} T^2 f^4 z$ has coefficient $-8\beta^{56} \neq 0$. This proves degree 99 for the actual fixed coefficients, not just for a synthetic coefficient example. The auxiliary residual is uniquely highest, of degree 28, so the sum of squares has degree 56.

For the 29-integer substitution give weight 2 to $J, F, \alpha, z, s, \eta, \zeta, w$ and weight 1 to the other old coordinates. The factor weights become

$$(44, 35, 50, 13, 4, 42),$$

whose sum is 188. The leading form (9) remains the weighted leader. Substitute each disjoint positive quadratic form, $w = u^2$, and the leading term $2r$ of $i = 2r + 1$. The result is nonzero: selecting the first square from every positive form and the $-zQ$ term gives coefficient $-32\beta^{56}$. The integer system has maximum degree 50 and its SOS degree 100. The analogous weighted leaders for normalized P_{67} and P_{68} give degrees 300 and 236, respectively. The full calculations are in `proofs/integer29_degrees.md`.

10 Verification, provenance, and limits of the claim

The archive is designed to run without network access. The arithmetic generators and auditors use the Python standard library. They verify the exact gate ledgers, absence of dead gates and free ports, agreement of factored and expanded quadratic relations, and the integer substitutions by exact arithmetic. Polynomial evaluations on affine lines supplement the uniform leading-form proofs; they do not replace them. Finite Pell checks test the identities and congruences used in the domain reductions, and the inherited local-window checks exercise actual marked halting control computations and invalid truncated runs. These tests do not exhibit a halting K215 run.

Every copied source has its repository path and SHA-256 in `SOURCE_MANIFEST.json`. The K215 table hash is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

The pinned endpoint README records Lean/Rocq checks for that same literal table. Those builds were not rerun for this report. No new theorem in either assistant is claimed. In particular the following evidence levels should remain distinct:

1. the existing external proof-assistant theorems for K215;
2. the ordinary K215 calibration and compiler proofs, formalizable in ZFC and used as mathematical dependencies here;
3. the new ordinary proofs in this report;
4. finite source, arithmetic, and layout audits.

An explicitly encoded derivation in ZFC's own proof syntax is not provided. Nevertheless, using ordinary mathematical proofs carried out within the resources of ZFC is the usual basis for the assertion "ZFC proves the equivalence." No assumption of Con(ZFC) is used in proving the equivalence itself.

The four coefficients are astronomically large. Their fixed recipes define actual integers, but the coefficient-height cost is not charged in the circuit metric. No binary expansion of $B = 2^{5^{67}}$, no enumeration of all approximately $1.42 \cdot 10^{21}$ windows, and no complete Diophantine witness is generated. A recipe of modest file size must not be confused with small explicit coefficient values.

If ZFC is consistent, it cannot prove the asserted absence of solutions, by the equivalence and the second incompleteness theorem. Mere consistency is insufficient to claim that ZFC also cannot prove the existence of a solution; such a two-sided independence assertion requires an additional suitable soundness assumption. This report proves neither consistency nor inconsistency.

11 Further research questions

1. **Formalize the small new interface.** The most focused target is the ordinary rank lemma, signed auxiliary target recovery, and the 29-integer substitution theorem. They can be proved parametrically before formalizing any astronomical fixed coefficient.
2. **Close the compiler formalization gap.** Connect an executable finite-window coefficient compiler and its carry/population proofs to the existing K215 endpoint. This would make the strongest current unformalized dependency explicit in a proof assistant.
3. **Reduce the seven remaining positive coordinates.** They are $J, F, \alpha, z, s, \eta, \zeta$. A valid reduction needs a proof that zeros outside the positive domain are harmless, or a completeness construction meeting a narrower polynomial image. Congruence observations alone are not enough.

4. **Improve the product finalizer.** The degree-99 product costs two more operations than the existing degree-155 product. Can the ordinary strong equality be combined into one unit factor with lower cost, without admitting a negative-unit branch?
5. **Lower the quartic variable count.** The current 33-variable lift is explicit and verified. Eliminating a register must preserve both quadratic degree and positivity of the inverse lift; removing a name while creating cubic products is not an improvement of this metric.
6. **Reduce coefficient height.** A direct compiler for the C222 calculus, a smaller local alphabet, or a different mask layout might matter much more than one arithmetic gate when coefficient bits are charged. A 214-state TM alone would not necessarily improve the arithmetic skeleton.
7. **Compare with explicit low-variable universal forms.** Instantiating a classical nine-variable or modern integer universal construction with a fully bound proof predicate would provide a useful alternative. Its degree, coefficient recipe, arithmetic circuit, and formalization assumptions should all be disclosed together.

A A closed specification of the four coefficients

This appendix retains package 06's fixed recipe. The accompanying `vendor/blank67/compiler_constants.py`, `vendor/blank67/blank_helical.py`, and `data/k215.tm` are exact source copies with hashes. No coefficient from another package is substituted.

A.1 The local alphabet and the two distinguished windows

K215 has working states $0, \dots, 214$, halt state 215, and two symbols. For tableau padding only, make state 215 absorbing without changing the transition that enters it. Ordinary cell codes are

$$2t + b \quad (b \in \{0, 1\},\ 0 \le t \le 216),$$

where $t = 0$ means headless and $t > 0$ means state $t - 1$. Add initial phases L, Q, R with codes 434,435,436, and boundaries V_b, H_b with codes 437,438. Their phase payloads are respectively blank/headless, blank/state 0, and blank/headless. Thus the alphabet size is $a_* = 439$.

Every 3×3 window is read in row-major order at time offsets $-1, 0, +1$. A center-valid window satisfies these local clauses:

1. Vertical-boundary bits agree between center and following-time cell. No two V_b are horizontally adjacent and no two H_b are vertically adjacent.
2. If neither center nor right neighbor is V_b, their horizontal-boundary bits agree.
3. An interior cell has an initial phase exactly when its predecessor is H_b. Adjacent phases are LL, LQ, QR, RR; a first interior cell next to V_b is L, and a last one is R.
4. A head immediately before a following H_b is in state 215. No head moves out through a side boundary.
5. If the following cell is interior, its payload is the exact deterministic three-neighbor K215 update. Multiple arriving heads are forbidden. Boundary neighbors act as blank headless cells.

The executable predicate fixes these clauses for boundary centers as well; its exact Python source is part of the coefficient definition. The first window is

$$(438, 438, 438, 434, 435, 436, 0, 1, 188),$$

the actual first transition around the initial head. The second is

$$(0, 438, 0, 0, 437, 438, 0, 437, 0).$$

It is center-valid but globally impossible: its predecessor has vertical bit 0 and its center vertical bit 1, violating the predicate at that predecessor. Enumerate these two first, then every other center-valid window lexicographically. The exact cardinality is

$$k_* = 1\,420\,903\,358\,357\,108\,792\,377.$$

It equals $38\,256\,607\,897 \cdot 439^4$; the four corners are unconstrained. The included cardinality checker verifies the five-cell cross count against the local predicate by a finite partition.

A.2 Numerical layout

The following fixed integers specify the layout completely:

Symbol	Value
ℓ	71
$j_{\max}$	710 451 679 178 554 395 461
$e_d = k_* + 12a_*$	1 420 903 358 357 108 797 645
$M = k_* + 15a_* + 1$	1 420 903 358 357 108 798 963
$E_{\max} = 27M$	38 364 390 675 641 937 572 001
H_*	72 466 071 276 212 548 748 431
T_1	149 194 852 627 496 423 892 873
T_2	302 652 415 330 064 174 180 879
b	5^{33}
L_*	5^{34}
$d = bL_*$	5^{67}

Define $A_* = 2^{71}$, $r_* = 2^b$, $B = r_*^{L_*} = 2^d$, and

$$\mu = A_* - 2 + \sum_{j=1}^{j_{\max}} A_*^j.$$

For $u, v \in \{0, 1, 2\}$ and $0 \le t < a_*$ put

$$e(u, v, t) = k_* + 3a_* + (2 - u)3a_* + (2 - v)a_* + t, \qquad j(u, v, t) = 1 + (3u + v)a_* + t.$$

Let $\mathcal{P}$ contain the selector positions $0, \dots, k_* - 1$, all positions $e(u, v, t)$, the single dummy position e_d, and the four anchors $M, 3M, 9M, 27M$. These positions are distinct. For the ith window w_i define

$$c_i = 1 + \sum_{u,v=0}^{2} A_*^{j(u,v,w_i[3u+v])} + \sum_{t=1}^{4} A_*^{9a_*+t}.$$

For payload positions put $c_{e(u,v,t)} = A_*^{j(u,v,t)}$; for the four anchors in their displayed order put $c_{e_t} = A_*^{9a_*+t}$; put $c_{e_d} = 0$. Now set

$$\begin{aligned}
D_C &= r_*^{3a_*} + r_*^{H_*+a_*} + r_*^{8M} + r_*^{24M} + \sum_{e\in\mathcal{P}} c_e(r_*^{T_1-e} + r_*^{T_2-e}),\\
D_R &= r_*^{H_*},\\
M_C &= B - 1 - \sum_{e\in\mathcal{P}\setminus\{1\}} r_*^{e} - 2r_*^{e_d},\\
M_F &= \mu(r_*^{T_1} + r_*^{T_2}) + \sum_{u=0}^{1}\sum_{v=0}^{2}\sum_{t=0}^{a_*-1} r_*^{e(u,v,t)}\\
&\quad + \sum_{u=0}^{2}\sum_{v=0}^{1}\sum_{t=0}^{a_*-1} r_*^{H_*+e(u,v,t)} + r_*^{9M} + r_*^{27M} + 4.
\end{aligned}$$

The four coefficients used everywhere in this report are exactly

$$\boxed{\beta = B - 1, \quad K = D_C + BD_R, \quad M_C, \quad M_F^* = M_F + B - 1.}$$

All summation ranges and exponents here are fixed finite integers. The source table, local predicate, ordering, and formulas specify unique integers even though expanding them is infeasible.

A.3 Why this recipe encodes a blank halting computation

The two clause bands impose local selector/payload consistency and overlap; the four anchors align the temporal shift with whole cells. The mask population is d, $M_C \equiv 2 \pmod 4$, and $M_F \equiv 4 \pmod 8$. The chosen spacing makes $2D_C - D_R \equiv 3 \pmod 5$, so dummy-bit tuning is possible without a coefficient-dependent correction bit.

A marked periodic valid tableau contains a finite strip between vertical boundaries. The initial phase graph has no cycle through Q, so boundaries occur on both sides of Start. Periodicity gives a later horizontal boundary, and the rectangle between successive boundaries follows K215 from one blank initial head to halt. Conversely a finite halting computation can be padded independently in width and height, including to powers of five. Helical seams cross time-independent vertical boundaries, and preserve every local clause. This supplies the canonical N used in the square-scale argument. Full inherited proofs, not just this synopsis, are included in `dependencies/blank_helical_proof.md` and the compiler notes.

B Reproduction and artifact map

From the extracted archive directory run

```
python3 src/reproduce.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The reproduction driver checks source hashes, rebuilds the arithmetic packets in a temporary directory, compares them with the delivered packets, runs the independent circuit and quadratic audits, and runs the selected finite Pell and compiler checks. It does not download dependencies, build proof assistants, or expand giant coefficients.

Path	Contents
`report.tex`, `report.pdf`	Complete manuscript and compiled PDF.
`src/`	Standard-library generators and independent verification driver.
`data/`	Literal K215 table; emitted positive, integer, and quartic specifications.
`proofs/`	Ordinary-root proof, signed-domain proof, independent sign review, exact integer degrees.
`vendor/blank67/`	Exact coefficient recipe, local predicate, and parent arithmetic packets.
`dependencies/`	Pinned ordinary mathematical proof dependencies and comparison record.
`SOURCE_MANIFEST.json`	Source paths, snapshot identities, and hashes.
`VERIFICATION.json`	Results and scope of checks performed for this delivery.
`licenses/`	Inherited repository and machine-source notices.

The deliverable is prepared for repository incorporation; no remote repository change is part of this archive delivery.

References

[1] ProveIt, *Con(ZFC) in Diophantine form*, eight-package reconciliation, snapshot `3b201cdea32d2c816f5f3d20734051f25fd3ef14`. Pinned repository collection.

[2] ProveIt, package 06, *Explicit Diophantine equations for Con(ZFC)*, `report.tex`, and its proof notes, 22 October 2026. Located in the collection of reference [1].

[3] ProveIt, package 02, *A fixed Diophantine representation of Con(ZFC)*, `report.tex` and `PELL_INTERFACE_LEMMAS.md`, 22 October 2026.

[4] ProveIt, package 08, *Explicit Diophantine equations for the consistency of ZFC*, and `start_only_quartic34_proof.md`, 22 October 2026.

[5] ProveIt, *Recovering integrality after relaxing an auxiliary Pell norm*, `Computability/HilbertTenthProblem/Papers/1980/PELL_RELAXED_AUXILIARY_PROOF.md`. Copied unchanged in this archive's `dependencies/`.

[6] ProveIt, `EXPLORATION_FIXED_MINUS_INDEX_PARITY.md`, `HALF_PARAMETER_PELL_92_PROOF.md`, and `complete75_reversed_auxiliary89.md`. Pinned source copies are in `dependencies/`.

[7] ProveIt, `pell_kernel_half_binomial42.md`, `complete75_asymmetric_scale_tradeoffs.md`, and `complete75_half_binomial_compiler.md`, under `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/`.

[8] ProveIt, *Formal correctness of the exact 215-state machine*, Pinned endpoint verification directory.

[9] J. P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic **47** (1982), 549–571, doi:10.2307/2273588. The repository includes a corrected annotated transcription.

[10] J. Bayer, M. David, M. Haßler, Y. Matiyasevich, and D. Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909v2 (2025). https://arxiv.org/abs/2506.20909.